

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

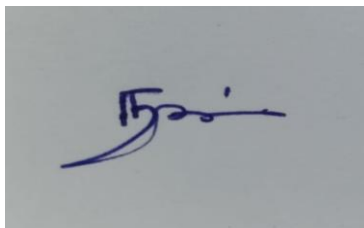
Total Marks: 50

Annexure A

Code of Conduct:

I MS NAVEEN 192371002 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink, appearing to read 'MS NAVEEN', is shown within a rectangular frame.

Annexure A Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given: $p = 17$, $q = 19$ Compute n and $\phi(n)$
Choose $e = 5$ and calculate the private key d Encrypt the message $M = 7$
Decrypt the ciphertext and verify correctness
Analyze the impact of choosing a small prime number on RSA security

2. ElGamal Encryption Analysis Given: $p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$
Compute the public key
Perform encryption to generate ciphertext pair $(C1, C2)$
Decrypt the ciphertext to retrieve the original message
Analyze how randomness (k) improves security in ElGamal

3. RSA Digital Signature Verification Given: $p = 11$, $q = 17$, $e = 7$, message $M = 8$
Compute n and $\phi(n)$
Find private key d
Generate the digital signature for the message
Verify the signature using the public key
Analyze how digital signatures ensure non-repudiation

4. Diffie-Hellman with Attack Resistance Evaluation Given:
 $p = 29$, $g = 2$ User A private key = 5
User B private key = 12
Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

5. Hybrid Encryption System Analysis Given:
A system uses RSA for key exchange and AES for data encryption
RSA encryption time = 5 ms per key exchange
AES encryption time = 0.5 ms per data block
Total data blocks = 500
Calculate total time taken for encryption
Compare with using only RSA for encrypting all data
Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure A – Analytical Problem Solving
Cryptography and Network Security

1. RSA Key Generation and Security Analysis
Given

- Prime number ($p = 17$)
 - Prime number ($q = 19$)
 - Public exponent ($e = 5$)
 - Message ($M = 7$)
-

Step 1: Compute n and $\phi(n)$

Formula

$$[$$

$$n = p \times q$$

$$]$$

$$[$$

$$n = 17 \times 19 = 323$$

$$]$$

Euler Totient:

$$[$$

$$\phi(n) = (p-1)(q-1)$$

$$]$$

$$[$$

$$= (17-1)(19-1)$$

$$]$$

$$[$$

$$= 16 \times 18 = 288$$

$$]$$

Answer

- $n = 323$
 - $\phi(n) = 288$
-

Step 2: Find Private Key (d)

Need

$$[$$

$$e \times d \equiv 1 \pmod{288}$$

$$]$$

Since

$$[$$

$$5 \times 173 = 865$$

$$]$$

$$\left[\begin{array}{l} 865 \div 288 = 3 \text{ remainder } 1 \end{array} \right]$$

Therefore

$$\left[\begin{array}{l} d=173 \end{array} \right]$$

Private Key = (173,323)

Step 3: Encrypt Message

Formula

$$\left[\begin{array}{l} C=M^e \bmod n \end{array} \right]$$

$$\left[\begin{array}{l} C=7^5 \bmod 323 \end{array} \right]$$

Calculate

$$\left[\begin{array}{l} 7^2=49 \end{array} \right]$$

$$\left[\begin{array}{l} 7^4=49^2=2401 \end{array} \right]$$

$$\left[\begin{array}{l} 2401 \bmod 323 = 140 \end{array} \right]$$

$$\left[\begin{array}{l} 7^5 = 140 \times 7 = 980 \end{array} \right]$$

$$\left[\begin{array}{l} 980 \bmod 323 = 11 \end{array} \right]$$

Ciphertext

[
 $\boxed{C=11}$
]

Step 4: Decrypt Ciphertext

Formula

[
 $M=C^d \bmod n$
]

[
 $M=11^{173} \bmod 323$
]

Using modular exponentiation,

[
 $11^{173} \bmod 323 = 7$
]

Recovered message

[
 $\boxed{M=7}$
]

Hence,

Encryption and decryption are verified successfully.

Security Analysis

Using small prime numbers makes RSA insecure because:

- n can be factorized easily.
 - Private key can be computed quickly.
 - Vulnerable to brute-force attacks.
 - Modern RSA uses 2048-bit or larger keys.
-

2. ElGamal Encryption Analysis

Given

- Prime ($p=23$)
- Generator ($g=5$)
- Private key ($x=7$)
- Random number ($k=3$)

- Message (M=10)

Step 1: Compute Public Key

Formula

$$[y = g^x \bmod p]$$

$$[= 5^7 \bmod 23]$$

Calculations

$$[5^2 = 25 \equiv 2]$$

$$[5^4 = 2^2 = 4]$$

$$[5^7 = 5^4 \times 5^2 \times 5]$$

$$[= 4 \times 2 \times 5 = 40]$$

$$[40 \bmod 23 = 17]$$

Public key

$$[\boxed{(p, g, y) = (23, 5, 17)}]$$

Step 2: Encryption

Compute

$$[C_1 = g^k \bmod p]$$

$$[$$

$$=5^3 \bmod 23$$

$$]$$

$$[$$

$$125 \bmod 23 = 10$$

$$]$$

Therefore

$$[$$

$$C_1 = 10$$

$$]$$

Compute

$$[$$

$$C_2 = M \times y^k \bmod p$$

$$]$$

First

$$[$$

$$17^2 = 289 \bmod 23 = 13$$

$$]$$

$$[$$

$$17^3 = 13 \times 17 = 221$$

$$]$$

$$[$$

$$221 \bmod 23 = 14$$

$$]$$

Now

$$[$$

$$C_2 = 10 \times 14 = 140$$

$$]$$

$$[$$

$$140 \bmod 23 = 2$$

$$]$$

Ciphertext

$$[$$

$$\boxed{(10, 2)}$$

$$]$$

Step 3: Decryption

Shared secret

$$[s = C_1^x \bmod p]$$

$$[= 10^7 \bmod 23]$$

Using modular exponentiation

$$[s = 14]$$

Inverse of 14 modulo 23

$$[14^{-1} = 5]$$

Recovered message

$$[M = C_2 \times s^{-1} \bmod 23]$$

$$[= 2 \times 5 = 10]$$

$$[10 \bmod 23 = 10]$$

Recovered message

$$[\boxed{M=10}]$$

Security Analysis

Random value k:

- Produces different ciphertexts for the same message.
- Prevents replay attacks.

- Protects against ciphertext analysis.
 - Makes ElGamal probabilistic and highly secure.
-

3. RSA Digital Signature Verification

Given

- $p = 11$
 - $q = 17$
 - $e = 7$
 - $M = 8$
-

Step 1: Compute n and $\phi(n)$

$$[$$

$$n = 11 \times 17 = 187$$

$$]$$

$$[$$

$$\phi(n) = 10 \times 16 = 160$$

$$]$$

Step 2: Find Private Key

Need

$$[$$

$$7d \equiv 1 \pmod{160}$$

$$]$$

$$[$$

$$7 \times 23 = 161$$

$$]$$

$$[$$

$$161 \pmod{160} = 1$$

$$]$$

Therefore

$$[$$

$$\boxed{d=23}$$

$$]$$

Step 3: Generate Signature

Formula

$$[$$

$$S = M^d \bmod n$$

$$]$$

$$[$$

$$S = 8^{23} \bmod 187$$

$$]$$

Using modular exponentiation

$$[$$

$$\boxed{S = 173}$$

$$]$$

Step 4: Verify Signature

Formula

$$[$$

$$M = S^e \bmod n$$

$$]$$

$$[$$

$$173^7 \bmod 187 = 8$$

$$]$$

Recovered message

$$[$$

$$\boxed{8}$$

$$]$$

Signature is valid.

Non-Repudiation Analysis

Digital signatures:

- Verify sender identity.
- Ensure message integrity.
- Prevent sender from denying the message.
- Widely used in e-commerce, banking and legal documents.

4. Diffie–Hellman Key Exchange

Given

- $p = 29$
- $g = 2$

- User A private key = 5
 - User B private key = 12
-

Step 1: Public Keys

User A

[
 $A=2^5 \bmod 29$
]

[
 $32 \bmod 29 = 3$
]

Public Key

[
 $\boxed{A=3}$
]

User B

[
 $B=2^{12} \bmod 29$
]

[
 $2^{12} = 4096$
]

[
 $4096 \bmod 29 = 7$
]

Public Key

[
 $\boxed{B=7}$
]

Step 2: Shared Secret

User A

[
 $K=7^5 \bmod 29$
]

```

[
=16
]
User B
[
 $K=3^{\{12\}} \bmod 29$ 
]
[
=16
]
Shared Secret
[
 $\boxed{K=16}$ 
]

```

Attack Analysis

If an attacker replaces public keys:

- User A establishes key with attacker.
- User B also establishes key with attacker.
- Attacker decrypts, modifies and re-encrypts messages.

This is called the Man-in-the-Middle (MITM) attack.

Solution

Use:

- Digital Signatures
- RSA Certificates
- Public Key Infrastructure (PKI)
- TLS/SSL authentication

These verify public keys before key exchange.

5. Hybrid Encryption System Analysis

Given

- RSA key exchange = 5 ms
 - AES encryption = 0.5 ms per block
 - Data blocks = 500
-

Step 1: Total Time (Hybrid System)

AES time

$$[500 \times 0.5 = 250 \text{ ms}]$$

Total

$$[250 + 5 = 255 \text{ ms}]$$

Hybrid Encryption Time

$$[\boxed{255 \text{ ms}}]$$

Step 2: RSA Only

RSA encrypts every block.

$$[500 \times 5 = 2500 \text{ ms}]$$

RSA Time

$$[\boxed{2500 \text{ ms}}]$$

Step 3: Comparison

Method	Time
--------	------

Hybrid (RSA + AES)	255 ms
--------------------	--------

RSA Only	2500 ms
----------	---------

Hybrid encryption is approximately

$$[\frac{2500}{255} \approx 9.8]$$

times faster.

Analysis

Why Hybrid Encryption is Efficient

- RSA encrypts only the AES session key.
 - AES encrypts bulk data very quickly.
 - Lower computation cost.
 - Better scalability for large files.
-

Justification in Modern Systems

Hybrid encryption is used because:

- Fast data encryption.
- Secure key exchange.
- Reduced computational overhead.
- High security with excellent performance.

It is used in:

- HTTPS
 - VPNs
 - Online Banking
 - Secure Email
 - Cloud Storage
-

Conclusion

The analytical problems demonstrate the practical implementation of major cryptographic algorithms:

- RSA provides secure encryption and digital signatures.
- ElGamal enhances security using random values during encryption.
- Diffie–Hellman enables secure secret-key exchange but requires authentication to prevent MITM attacks.
- Hybrid Encryption (RSA + AES) combines the strengths of asymmetric and symmetric cryptography, offering both strong security and high efficiency. These techniques form the foundation of modern secure communication systems used across banking, e-commerce, cloud computing, and internet security.

